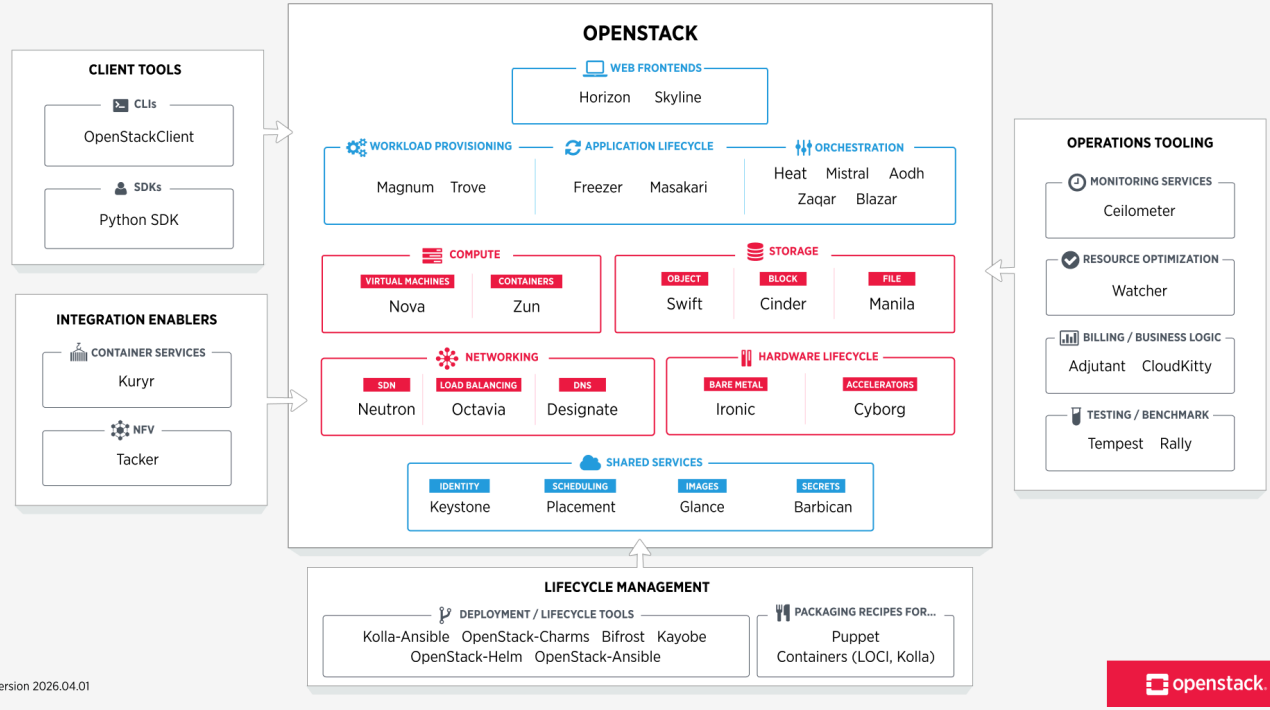
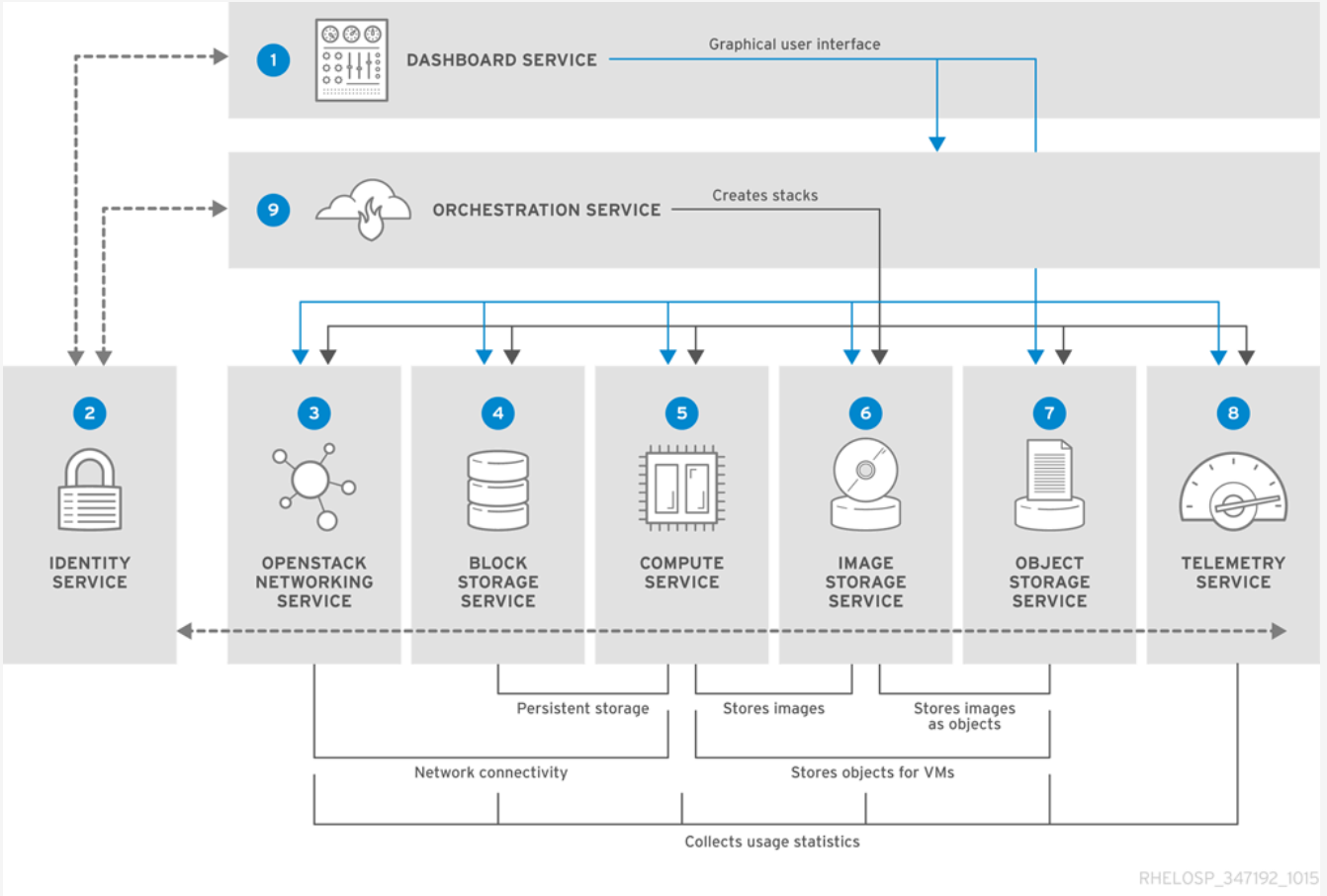
OpenStack

Architecture

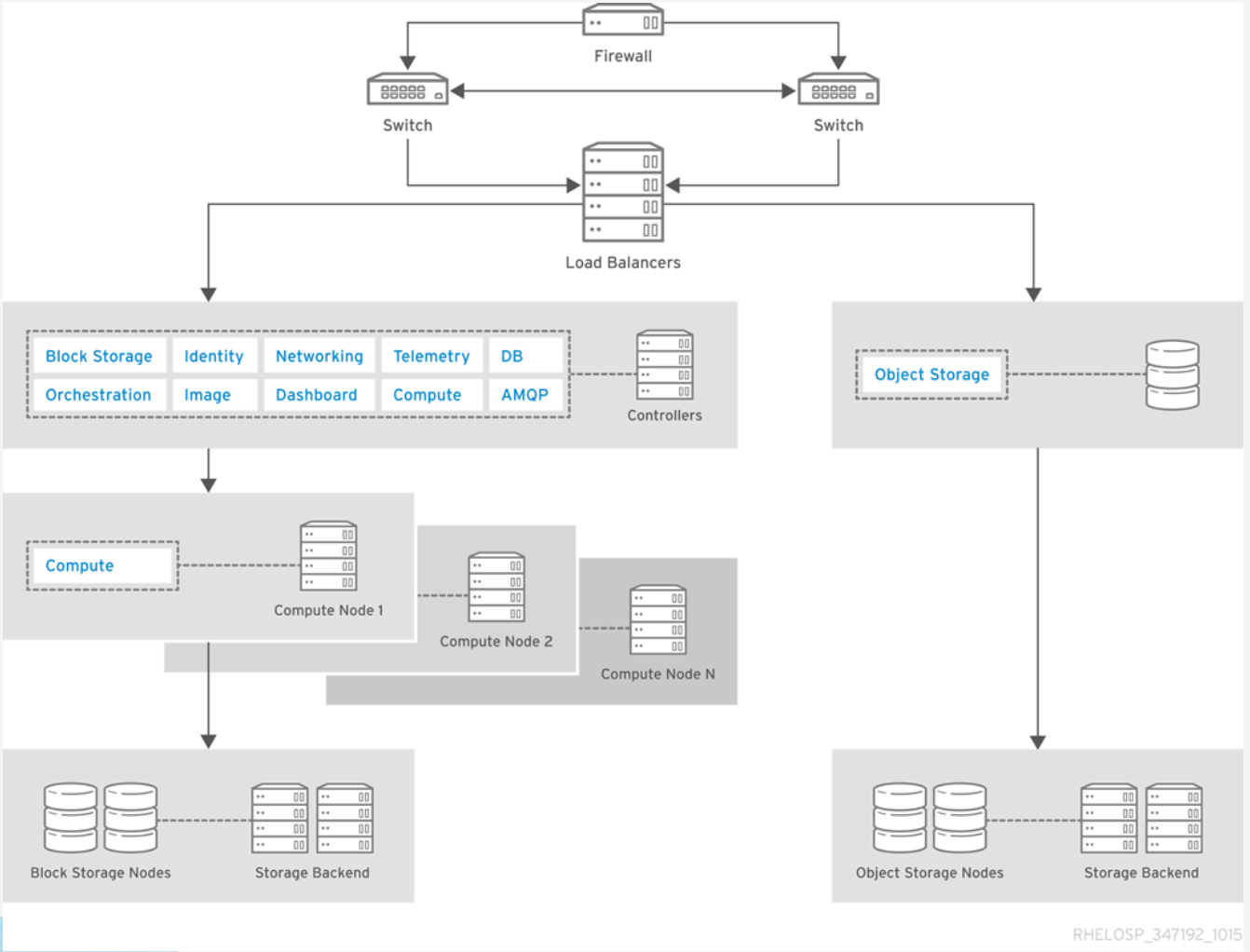


Version 2026.04.01

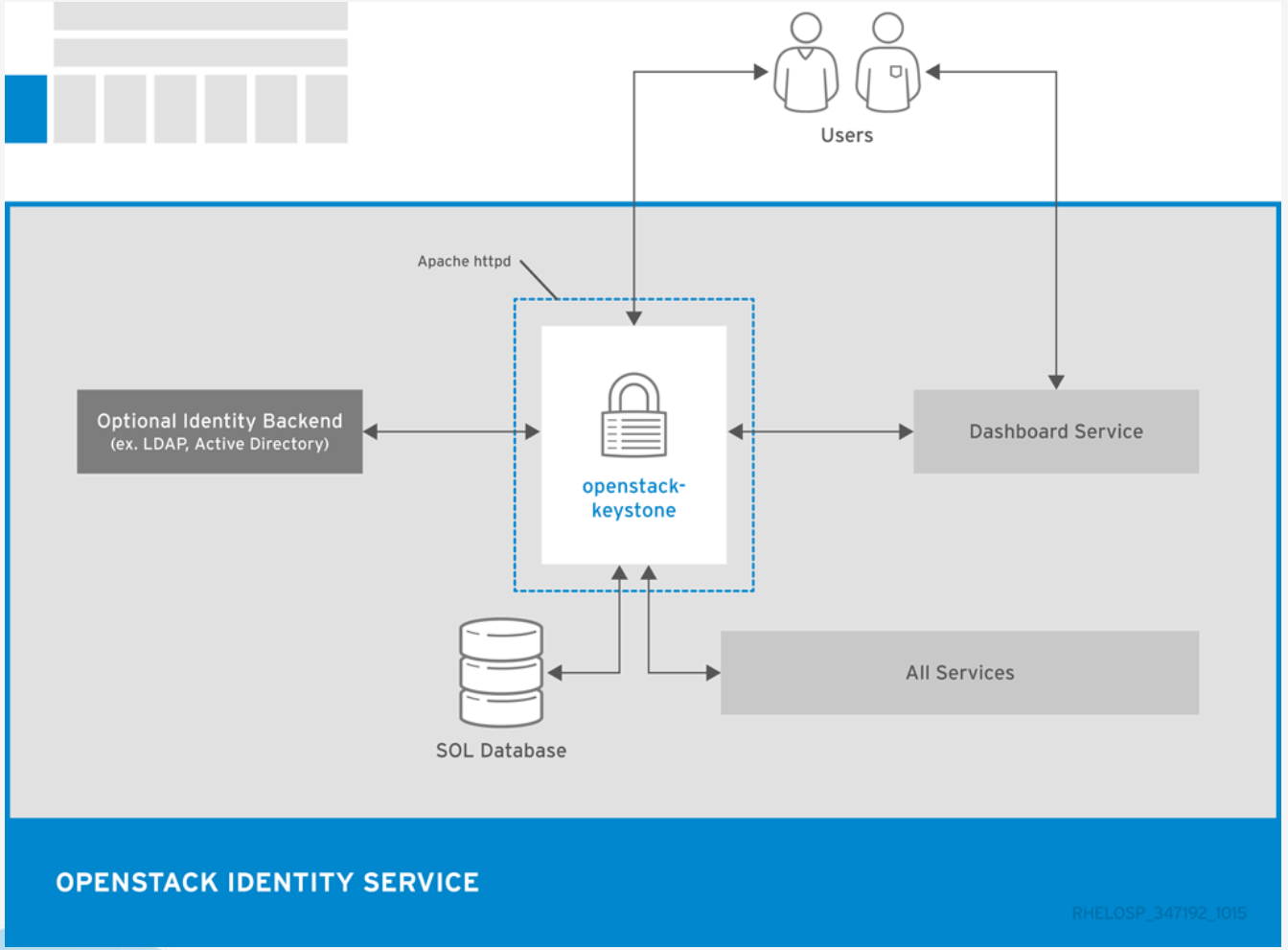
Architecture



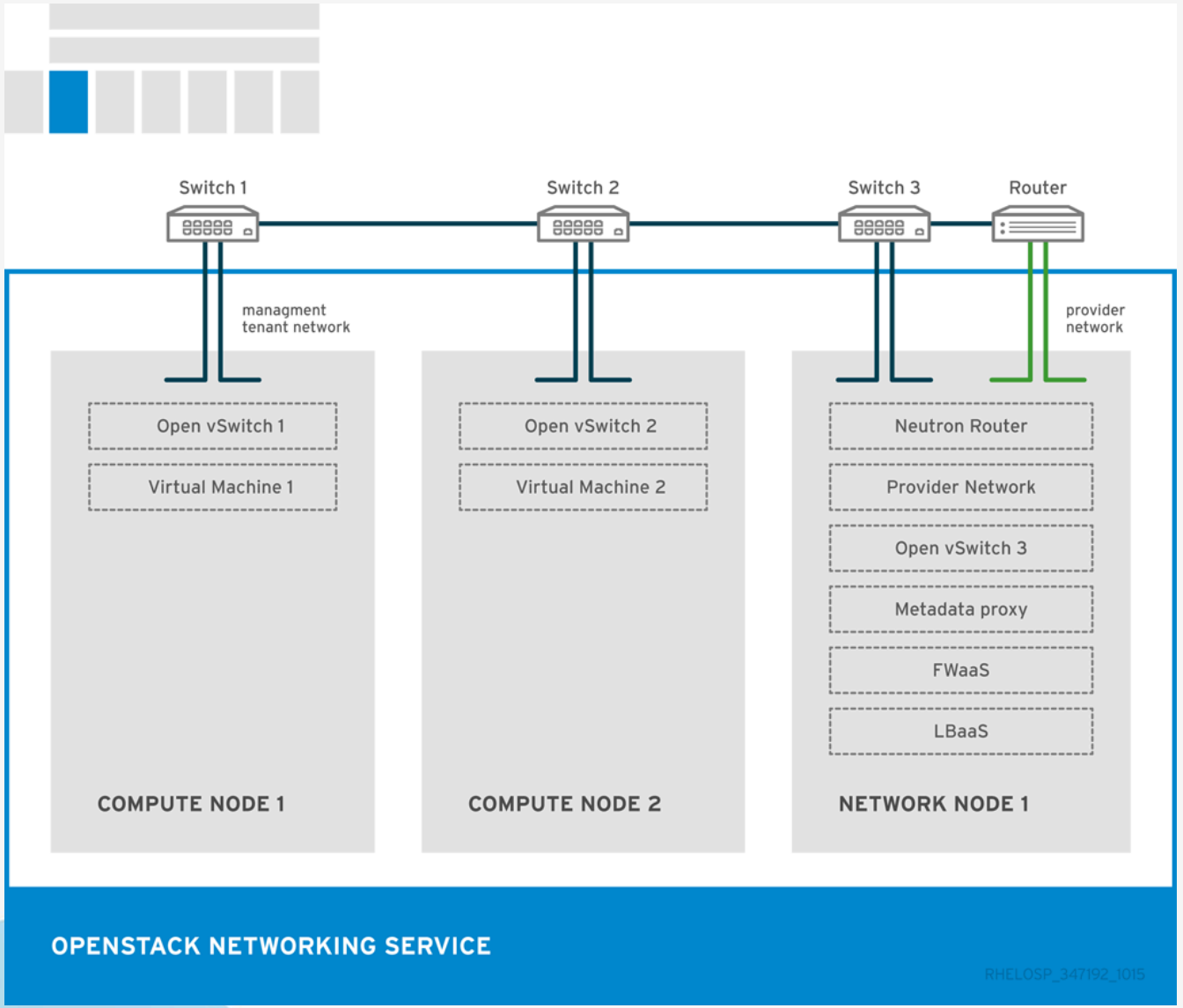
Architecture



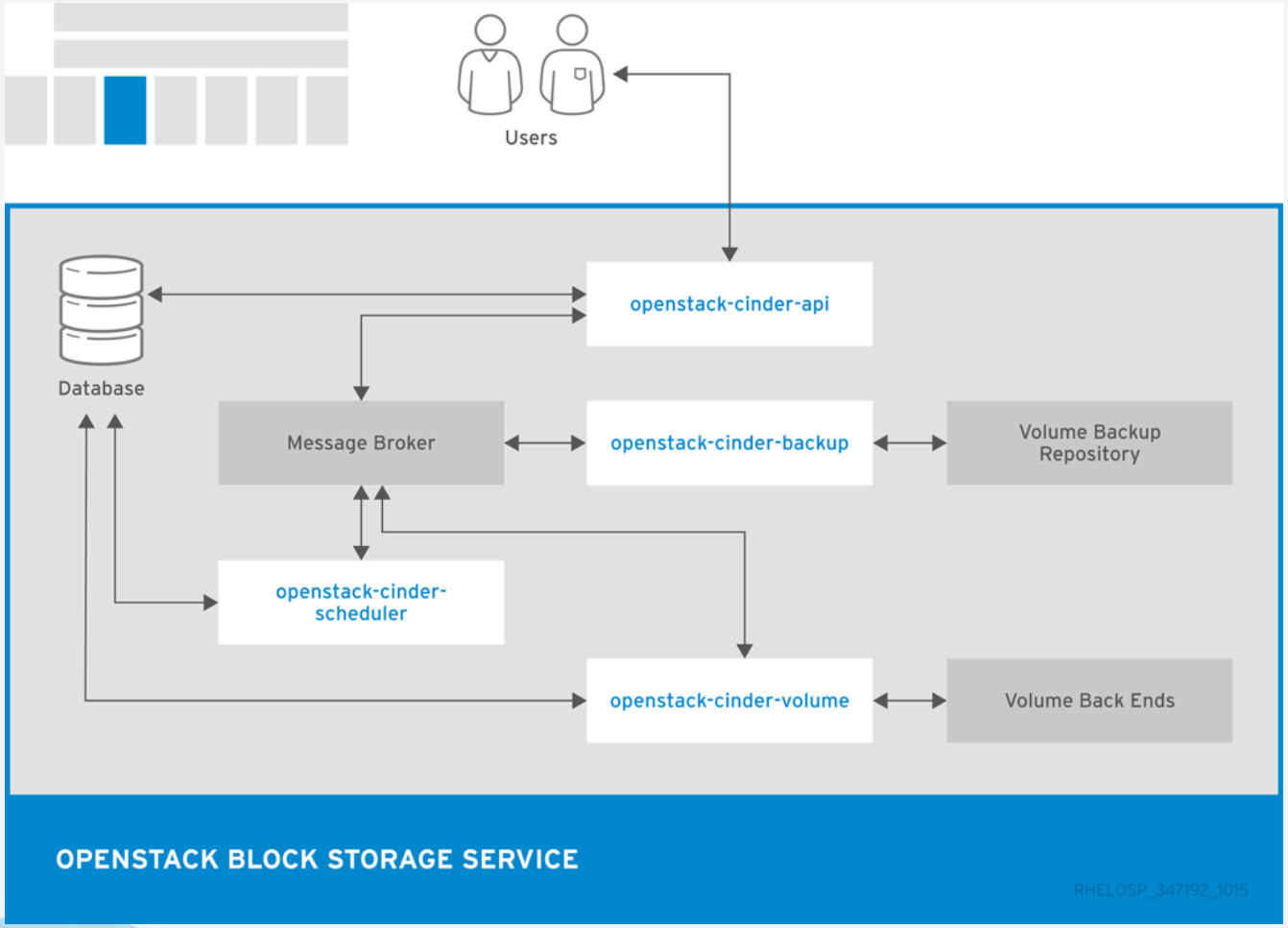
Architecture



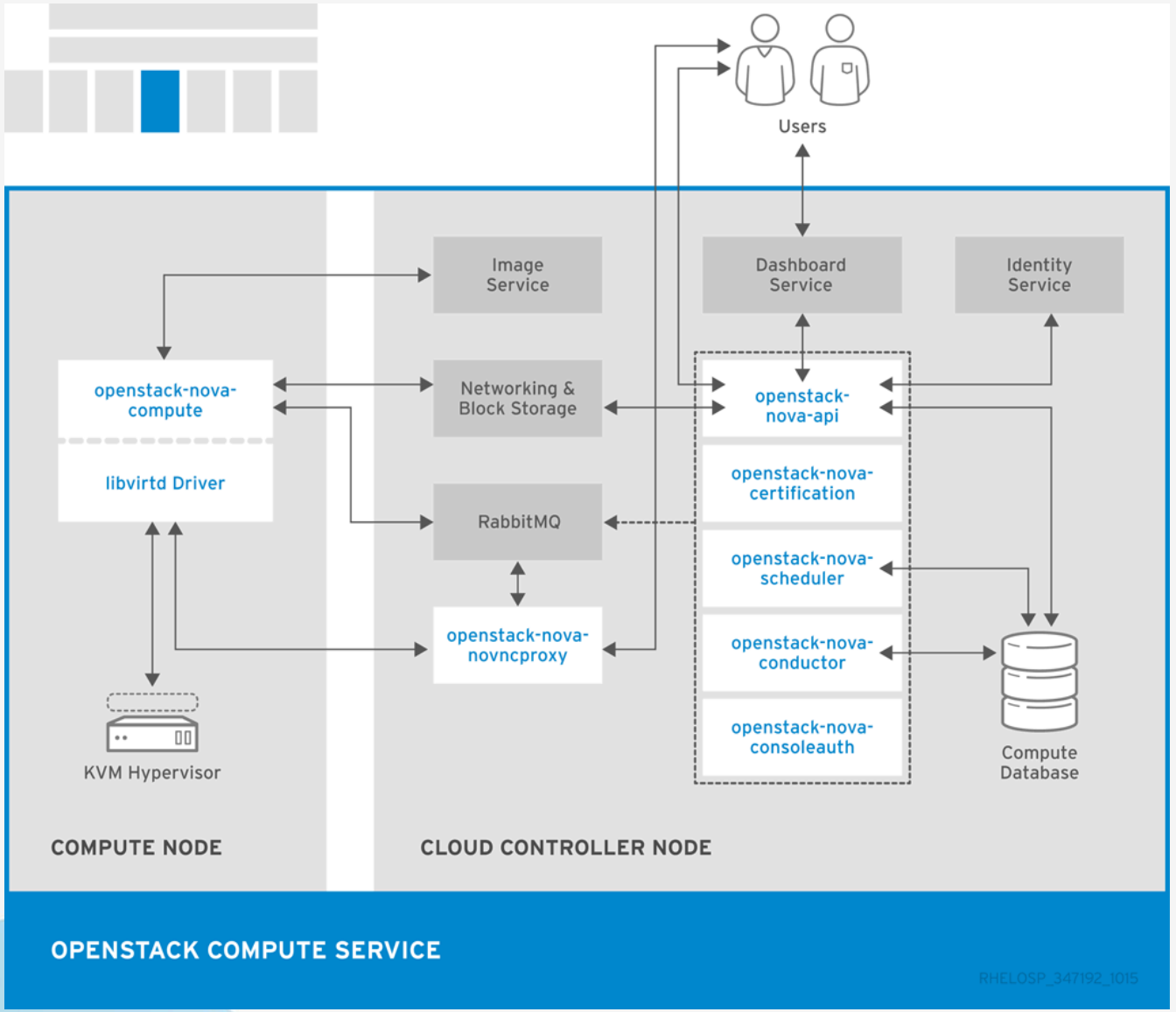
Architecture



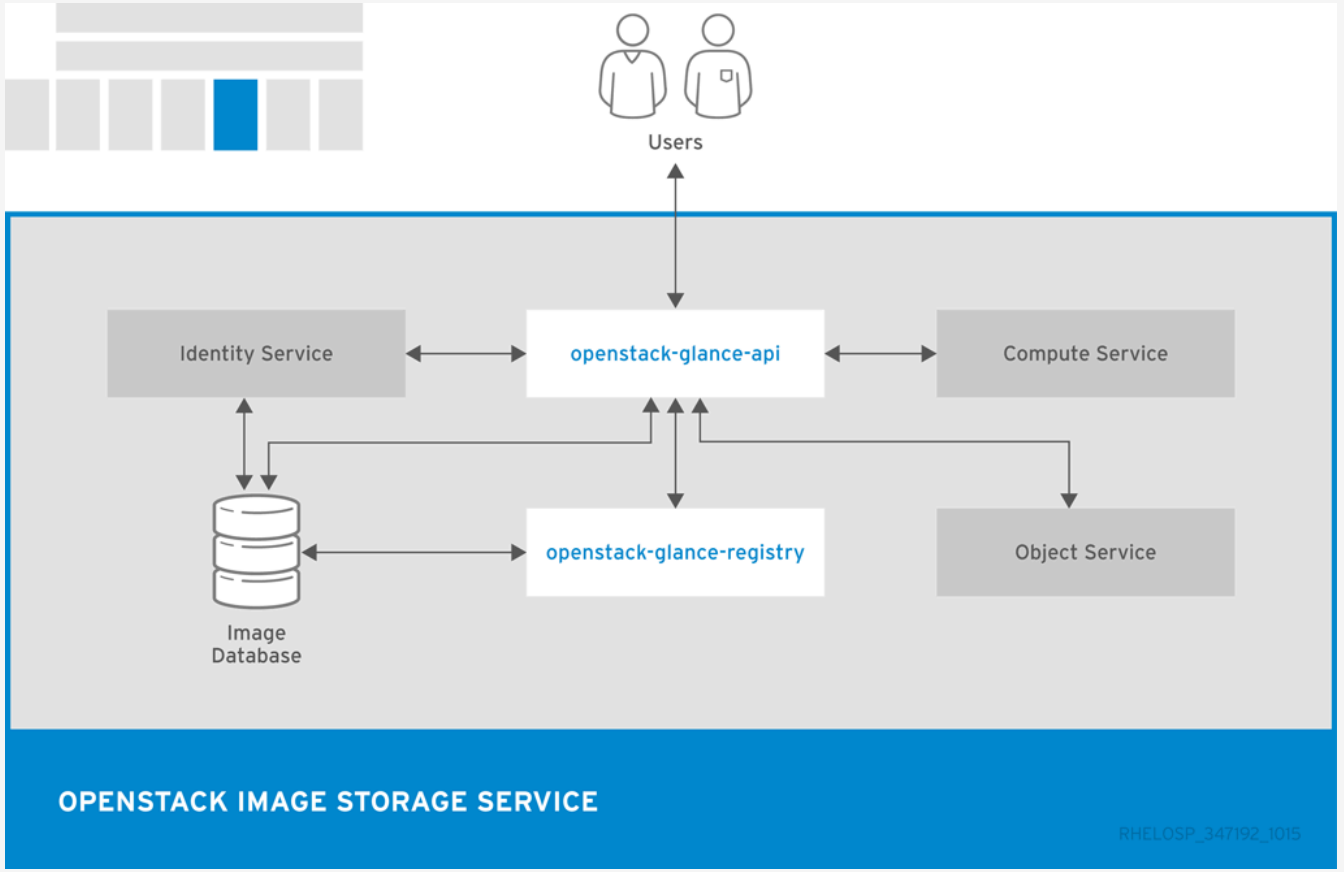
Architecture



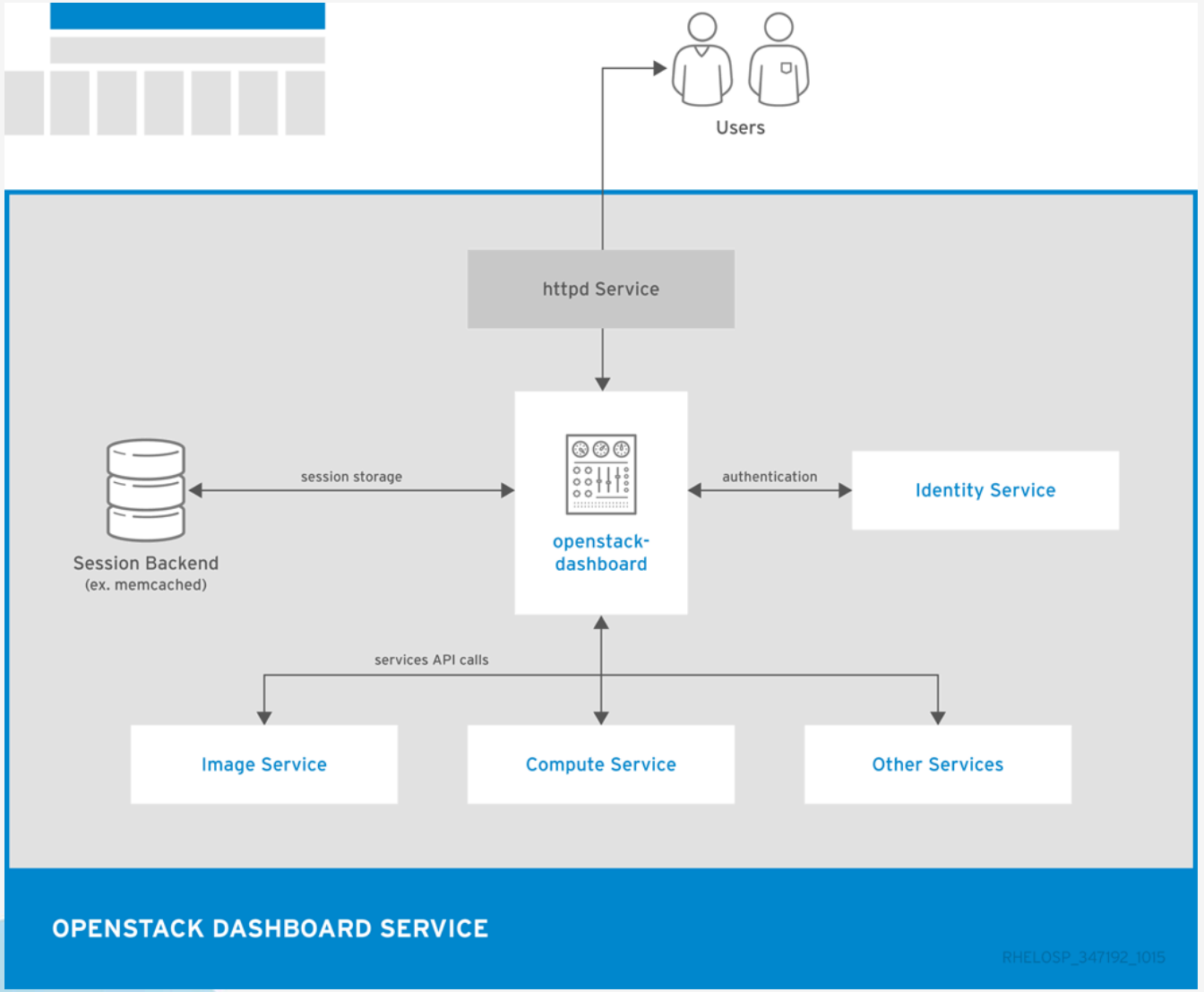
Architecture



Architecture

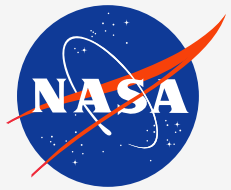


Architecture



Users


Walmart


NASA


HUAWEI

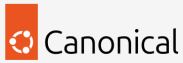

BLIZZARD


CERN


Deutsche
Telekom


Bloomberg

Contributors



Deploy OpenStack

KOLLA

an OpenStack Community Project

Deploy OpenStack

```
# Prepare Python virtual environment
cd ~/openstack/

python3 -m venv $VENV_PATH

ln -s  $VENV_PATH ~/venv/openstack
source $VENV_PATH/bin/activate

# Install needed pip packages
pip install -U pip
pip install -r requirements.txt

# Install dependencies for kolla-ansible
kolla-ansible install-deps

# ⚠️ NOTE: fix for prometheus 05.2026
# pip install "bcrypt<5.0.0"

# Install additional dependencies
gilt overlay

# Reload environment
source ~/.profile
```

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Deploy OpenStack

```
# Prepare ansible inventory

# Copy inventory file
cp ~/venv/openstack/share/kolla-ansible/ansible/inventory/multinode inventory/

# Patch inventory file and review inventory/wrx/ directory afterwards
crudini --del inventory/multinode control
crudini --del inventory/multinode network
crudini --del inventory/multinode compute
crudini --del inventory/multinode monitoring
crudini --del inventory/multinode storage
crudini --del inventory/multinode loadbalancer:children

# Prepare passwords
cp ~/venv/openstack/share/kolla-ansible/etc_examples/kolla/passwords.yml \
  custom-config/wrx/passwords.yml

kolla-genpwd -p custom-config/wrx/passwords.yml
```

KOLLA

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Deploy OpenStack

```
# Overview of kolla-ansible commands

kolla-ansible bootstrap-servers    -i inventory/wrx
kolla-ansible octavia-certificates -i inventory/wrx
kolla-ansible prechecks            -i inventory/wrx
kolla-ansible pull                 -i inventory/wrx
kolla-ansible deploy               -i inventory/wrx  -vvvv
kolla-ansible post-deploy          -i inventory/wrx
kolla-ansible check                -i inventory/wrx
kolla-ansible destroy              -i inventory/wrx <--yes-i-really-really-mean-it>
kolla-ansible mariadb_recovery     -i inventory/wrx
kolla-ansible reconfigure          -i inventory/wrx
kolla-ansible upgrade              -i inventory/wrx
kolla-ansible stop                 -i inventory/wrx
kolla-ansible deploy-containers    -i inventory/wrx
kolla-ansible prune-images         -i inventory/wrx
```

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Deploy OpenStack

```
# Prepare the kolla-ansible deployment

kolla-ansible bootstrap-servers    -i inventory/wrx
kolla-ansible octavia-certificates -i inventory/wrx
kolla-ansible prechecks            -i inventory/wrx

# ⚠️ NOTE: this will take time
kolla-ansible pull                 -i inventory/wrx
```

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Deploy OpenStack

```
# Prepare the kolla-ansible deployment

# ⚠️ NOTE: this will take time
kolla-ansible deploy               -i inventory/wrx

# Post deployment steps
kolla-ansible post-deploy          -i inventory/wrx
kolla-ansible check                -i inventory/wrx
```

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Bootstrap a Test Environment

```
# Create first basic test resources
KOLLA_CONFIG_PATH=~/openstack/custom-config/wrx/    \
ENABLE_EXT_NET=1                                   \
EXT_NET_CIDR=10.30.0.0/24                          \
EXT_NET_GATEWAY=10.30.0.1                          \
EXT_NET_RANGE="start=10.30.0.240,end=10.30.0.245"  \
DEMO_NET_DNS=10.30.0.1                             \
~/venv/openstack/share/kolla-ansible/init-runonce
```

KOLLA

an OpenStack Community Project

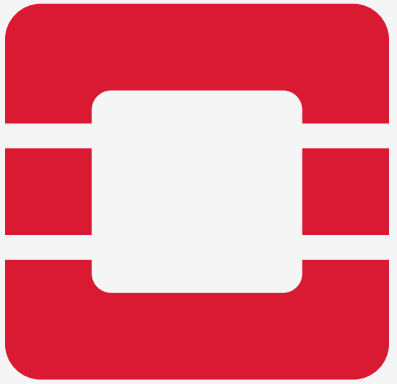
Access

Horizon	http://int.os.wrx.sckt.net
Skyline	http://int.os.wrx.sckt.net:9999
CLI	Command Line Interface

Check the following files to use with CLI:

- custom-config/wrx/admin-openrc.sh
- custom-config/wrx/clouds.yaml

Operations



Service Overview

```
# Review the OpenStack environment
openstack service  list
openstack endpoint list
openstack catalog  list

openstack compute       service list
openstack volume        service list
openstack network       agent   list
openstack orchestration service list
```

Replace a Network Node

```
# 🔍 Review data stored on the node. Is there anything to back up?
docker volume ls
du -lhs /var/lib/docker/volumes/*

# 💥 Break the network03 node
apt remove  docker*  -y
rm -rf  /etc/kolla /var/log/kolla  \
        /etc/systemd/system/kolla* \
        /etc/systemd/system/multi-user.target.wants/kolla-* \
        /etc/docker/daemon.json
umount /var/lib/docker;  wipefs /dev/sdb -a; mkfs.ext4 /dev/sdb
reboot

# 🚚 Redeploy OpenStack services on network03 from the deployment node
kolla-ansible bootstrap-servers -i inventory/wrx --limit ~network03
kolla-ansible prechecks         -i inventory/wrx --limit ~network03
kolla-ansible pull              -i inventory/wrx --limit ~network03
kolla-ansible reconfigure       -i inventory/wrx --limit ~network03
```

Replace a Controller Node

```
# 🔍 Review data stored on the node. Is there anything to back up?          📋
docker volume ls
du -lhs /var/lib/docker/volumes/*

# 💥 Break the controller03 node
systemctl stop kolla-* docker.service docker.socket
apt remove  docker*  -y
rm -rf  /etc/kolla /var/log/kolla  \
        /etc/systemd/system/kolla* \
        /etc/systemd/system/multi-user.target.wants/kolla-* \
        /etc/docker/daemon.json
umount /var/lib/docker;  wipefs /dev/sdb -a; mkfs.ext4 /dev/sdb
reboot

# 🚚 Redeploy OpenStack services on controller03 from the deployment node
kolla-ansible bootstrap-servers -i inventory/wrx --limit ~controller03
kolla-ansible prechecks         -i inventory/wrx --limit ~controller03
kolla-ansible pull              -i inventory/wrx --limit ~controller03

# ☕ this will take a while
kolla-ansible reconfigure       -i inventory/wrx --limit ~controller03 --skip-tags keystone

# ⚠️ we have to reconfigure keystone across all controller nodes ⚠️
kolla-ansible reconfigure       -i inventory/wrx                       --tags      keystone
```

Replace a Compute Node

```
# 🔍 Review data stored on the node. Is there anything to back up?          📋
docker volume ls
du -lhs /var/lib/docker/volumes/*
docker exec -ti nova_libvirt virsh list  --all

# 🚧 Drain the OpenStack compute host
openstack server list --all       --host    <COMPUTE_NAME>
openstack compute service set     --disable <COMPUTE_NAME> nova-compute
for vm in $(openstack server list --host    <COMPUTE_NAME> --all-projects -c ID -f value);
do
  echo "Evacuating VM: $vm"
  openstack server migrate --live $vm
done
openstack server list --all       --host    <COMPUTE_NAME>

# 💥 Break the node
# On the node
systemctl stop kolla-* docker.service docker.socket
apt remove  docker*  -y
rm -rf  /etc/kolla /var/log/kolla  \
        /etc/systemd/system/kolla* \
        /etc/systemd/system/multi-user.target.wants/kolla-* \
        /etc/docker/daemon.json
cp -pr  /var/lib/docker/volumes ~/
umount  /var/lib/docker;  wipefs /dev/sdb -a; mkfs.ext4 /dev/sdb
reboot

# 🚚 Redeploy OpenStack services on compute03 from the deployment node
kolla-ansible bootstrap-servers -i inventory/wrx --limit ~<COMPUTE_NAME>
kolla-ansible prechecks         -i inventory/wrx --limit ~<COMPUTE_NAME>
kolla-ansible pull              -i inventory/wrx --limit ~<COMPUTE_NAME>
kolla-ansible reconfigure       -i inventory/wrx --limit ~<COMPUTE_NAME>
```

Debugging

```
# 🩺 Check                                                                    📋
openstack compute service list

# 🔍 Take a look at logs on compute03
tail -f /var/log/kolla/nova/nova-compute.log

# 🔍 Compare files on nova_compute volume and in the backup
ls -lh ~/volumes/nova_compute/_data/ /var/lib/docker/volumes/nova_compute/_data/

# ⚙️ Repair
cp ~/volumes/nova_compute/_data/compute_id  /var/lib/docker/volumes/nova_compute/_data/

# 🩺 Check the status again and test
openstack compute service list
openstack compute service set --enable compute03 nova-compute
openstack server migrate --live --host compute03 <SERVER_ID>
openstack server list    --all  --long
```

Scale-out

- How to scale out controllers?
- How to scale out network nodes?
- How to scale out compute nodes?

Add a New Compute Host

```
# 📝 Modify inventory, put the new compute host into the hypervisor section 📋
vim inventory/wrx/20_openstack

# 🩺 Check the new host

# 🚚 Deploy OpenStack services on the new compute host
kolla-ansible bootstrap-servers -i inventory/wrx --limit ~compute12
kolla-ansible prechecks         -i inventory/wrx --limit ~compute12
kolla-ansible pull              -i inventory/wrx --limit ~compute12
kolla-ansible deploy            -i inventory/wrx --limit ~compute12

# 🩺 Check the new host
openstack compute service list
openstack server  migrate --live --host compute12 <SERVER_ID>

docker exec -ti nova_libvirt virsh list  --all # on the host
```

Remove a Compute Host

Reboot an OpenStack Host

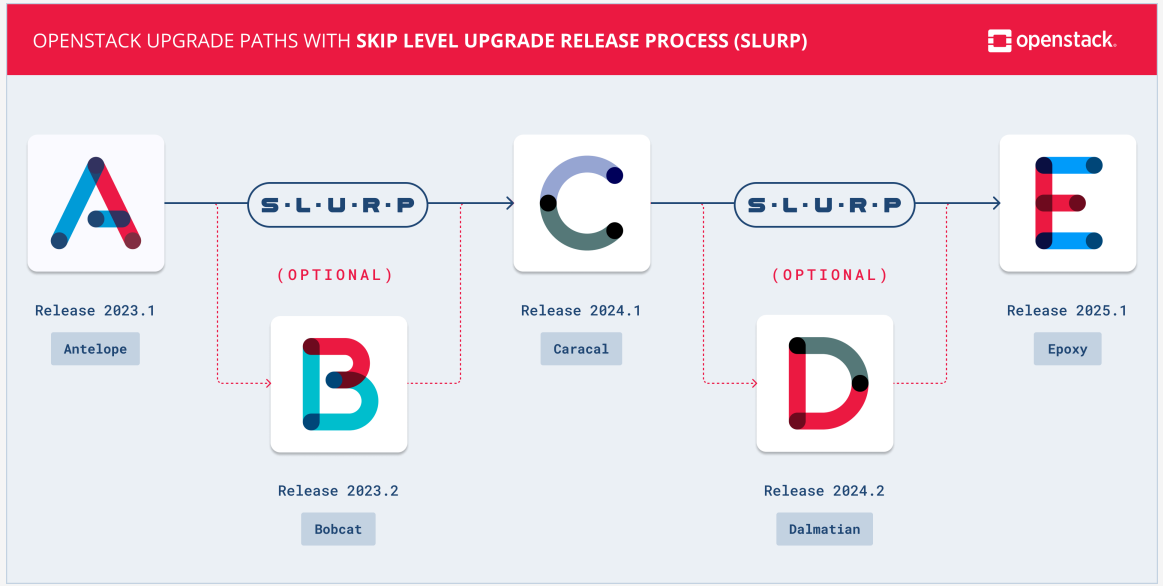
- How to reboot controllers?
- How to reboot network nodes?
- How to reboot compute nodes?

Debugging RabbitMQ

```
# Some commands to debug RabbitMQ errors

rabbitmqctl status
rabbitmqctl cluster_status

rabbitmq-plugins list
rabbitmq-plugins disable <PLUGIN>
rabbitmq-plugins enable  <PLUGIN>

rabbitmqctl list_queues name,messages,messages_ready,messages_unacknowledged
rabbitmqctl list_bindings

rabbitmqctl purge_queue   <QUEUE_NAME>
rabbitmqctl delete_queue  <QUEUE_NAME>
```

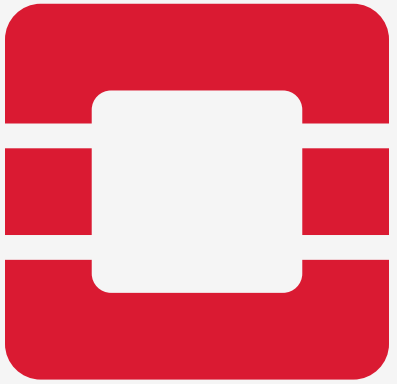
Upgrade

SLURP Model


OPENSTACK UPGRADE PATHS WITH SKIP LEVEL UPGRADE RELEASE PROCESS (SLURP)
openstack.
S·L·U·R·P
(OPTIONAL)
S·L·U·R·P
(OPTIONAL)
Release 2023.1
Antelope
Release 2023.2
Bobcat
Release 2024.1
Caracal
Release 2024.2
Dalmatian
Release 2025.1
Epoxy

Roll out the Next Release

```
# 📝 Update kolla-ansible                                                    📋
CUR_KA_VER=$(pip list | grep kolla | awk '{print $NF}' | awk -F . '{print $1}')

echo $VIRTUAL_ENV
deactivate
python3  -m venv ~/venv/kolla-ansible-$((CUR_KA_VER+1))
source  ~/venv/kolla-ansible-$((CUR_KA_VER+1))/bin/activate

pip install -U pip
pip install -r requirements.txt
pip install -U "kolla-ansible<$((CUR_KA_VER + 2)).0"
kolla-ansible install-deps
# ⚠️ NOTE: fix for prometheus 05.2026
# pip install "bcrypt<5.0.0"
```



Roll out the Next Release

Inventory

```
# Update inventory
mv inventory/multinode inventory/multinode.old
cp $VIRTUAL_ENV/share/kolla-ansible/ansible/inventory/multinode inventory/
crudini --del inventory/multinode control
crudini --del inventory/multinode network
crudini --del inventory/multinode compute
crudini --del inventory/multinode monitoring
crudini --del inventory/multinode storage
crudini --del inventory/multinode loadbalancer:children

# Review the differences
diff -y --suppress-common-lines  inventory/multinode  inventory/multinode.old
```

Roll out the Next Release

Passwords

```
# Update passwords
mv custom-config/wrx/passwords.yml custom-config/wrx/passwords.yml.old
cp $VIRTUAL_ENV/share/kolla-ansible/etc_examples/kolla/passwords.yml \
    custom-config/wrx/passwords.yml
kolla-genpwd -p custom-config/wrx/passwords.yml
kolla-mergepwd \
  --old   custom-config/wrx/passwords.yml.old \
  --new   custom-config/wrx/passwords.yml     \
  --final custom-config/wrx/passwords.yml

# Review the differences
diff -y --suppress-common-lines  custom-config/wrx/passwords.yml custom-config/wrx/passwords.yml.old
```

Roll out the Next Release

Go

```
# 💾 Create config backups                                                  📋
ansible  all -b -m ansible.builtin.shell -a \
  "mkdir -p ~/backup; cp  -r /etc/kolla ~/backup/$(date +%y%m%d%H%M%S)_kolla"

# 🗂️ Force to collect facts
kolla-ansible gather-facts -i inventory/wrx -e "ansible_python_interpreter=/usr/bin/python3"

# ansible all -m ansible.builtin.setup

# 🩺 Check
kolla-ansible prechecks    -i inventory/wrx

# 📥 Pull images
kolla-ansible pull         -i inventory/wrx

# 🚚 Upgrade
kolla-ansible upgrade      -i inventory/wrx
```

High Availability

- Keepalived & HAProxy (controllers)
- Stateless API Services
- MariaDB Galera Cluster as State Store
- Message Broker (RabbitMQ)
- Multiple nodes of each type

Debugging Common

```
# Some useful commands for debugging

# On each node
docker ps  | grep unhealthy
df -Ph

# On controller node
tail -f /var/log/kolla/mariadb/mariadb.log
tail -f /var/log/kolla/rabbitmq/rabbit*.log

# On deployment node
openstack compute service list
openstack volume  service list
openstack network agent   list

# Check HA IP on controller nodes
ip -br a

# Search for error messages connected with a resource
grep -r [RESOURCE_ID] /var/log/kolla/[SERVICE]
```

Backup and Recovery

- MySQL Backup
- Config Backup (can be restored by kolla)
- Volumes Backup (can be ⚠️ partially restored)
- Message Queues (are recreated if lost)

Database Backup and Recovery

```
# Create a new backup
kolla-ansible mariadb-backup -i inventory/wrx

# Check on the controller
ls -lh /var/lib/docker/volumes/mariadb_backup/_data/

# To restore
gunzip -c <BACKUP_FILE> | docker exec -i mariadb mysql -u root -p

# If the DB cluster cannot create a quorum
kolla-ansible mariadb-recovery -i inventory/wrx
```

Monitoring

- Ceilometer / Gnocchi / Aodh (for Billing)
- Prometheus / Alertmanager / Grafana

Grafana	http://int.os.wrx.sckt.net:3000
Prometheus	http://int.os.wrx.sckt.net:9091
Alertmanager	http://int.os.wrx.sckt.net:9095

Expand / Deploy a New Service

Central Logging

```
# 🔌 Enable central logging                                          📋
cat custom-config/wrx/globals.d/logging.yml
sed -e 's/enable_central_logging:.*/enable_central_logging: "yes"/' \
    -i custom-config/wrx/globals.d/logging.yml
cat custom-config/wrx/globals.d/logging.yml

# 🚚 Deploy the new service
kolla-ansible pull          -i inventory/wrx --tags opensearch
kolla-ansible deploy        -i inventory/wrx --tags opensearch
kolla-ansible reconfigure   -i inventory/wrx --tags common
```

Visit:

- http://int.os.wrx.sckt.net:5601
- visualize, create an index pattern
- discover

Using OpenStack

Images

```
# Prepare images
mkdir -p  ~/cloud-images/

ALMA_VERSION="10.2"
ALMA_IMG_NAME=$(
  curl -sS https://repo.almalinux.org/almalinux/10/cloud/x86_64/images/CHECKSUM  \
   | grep GenericCloud-$ALMA_VERSION | awk '{print $2}' |  sort  | tail -n1)
ALMA_LINK=https://repo.almalinux.org/almalinux/10/cloud/x86_64/images/$ALMA_IMG_NAME

ROCKY_VERSION="10.2"
ROCKY_IMG_NAME=$(
  curl -sS https://dl.rockylinux.org/pub/rocky/10/images/x86_64/CHECKSUM \
  | grep SHA256 | awk '{print $2}'  | grep GenericCloud-LVM-$ROCKY_VERSION \
  | sed -e "s/[()]//g" | sort | tail -n1)
ROCKY_LINK=https://dl.rockylinux.org/pub/rocky/10/images/x86_64/$ROCKY_IMG_NAME

UBUNTU_VERSION="24.04"
UBUNTU_IMG_NAME=noble-server-cloudimg-amd64.img
UBUNTU_LINK="https://cloud-images.ubuntu.com/noble/current/$UBUNTU_IMG_NAME"

DEBIAN_VERSION="13"
DEBIAN_IMG_NAME=debian-13-generic-amd64.qcow2
DEBIAN_LINK="https://cloud.debian.org/images/cloud/trixie/latest/$DEBIAN_IMG_NAME"
```

Images

```
# Download images
for i in  alma~$ALMA_VERSION~$ALMA_LINK       \
          rocky~$ROCKY_VERSION~$ROCKY_LINK    \
          ubuntu~$UBUNTU_VERSION~$UBUNTU_LINK \
          debian~$DEBIAN_VERSION~$DEBIAN_LINK
do
  IFS="~" read -r -a fields <<< $i
  name=${fields[0]}
  version=${fields[1]}
  link=${fields[2]}
  filepath=~/cloud-images/$(basename $link)

  echo "Downloading $name-$version image"
  curl --output-dir ~/cloud-images -L -O $link

  echo "Uploading $name-$version image into OpenStack"
  openstack image create "$name-$version"       \
      --file             "$filepath"            \
      --disk-format       qcow2                 \
      --container-format  bare                  \
      --property          os_distro="$name"     \
      --property          os_version="$version" \
      --public
done

# Check
openstack image list --long
```

Authentication / Authorization

```
# Set permissions
PRJ_NAME=workshop
USR_NAME=workshop
USR_PASS=workshop

# Create project
openstack project create                              $PRJ_NAME
openstack quota set --instances 32 --cores 64         $PRJ_NAME

# Create user
openstack user create --password $USR_PASS --project $PRJ_NAME $USR_NAME

# Create permissions for the user
openstack role add    --user $USR_NAME     --project $PRJ_NAME member

# Check
openstack role assignment list --name
openstack quota show $PRJ_NAME
```

How to authenticate with CLI

```
# Prepare an openrc file to be used with the OpenStack CLI
cp  custom-config/wrx/admin-openrc.sh  custom-config/wrx/workshop-openrc.sh

sed  -e "s/OS_PROJECT_NAME=.*/OS_PROJECT_NAME=$PRJ_NAME/" -i custom-config/wrx/workshop-openrc.sh
sed  -e "s/OS_TENANT_NAME=.*/OS_TENANT_NAME=$PRJ_NAME/"   -i custom-config/wrx/workshop-openrc.sh
sed  -e "s/OS_PASSWORD=.*/OS_PASSWORD=$USR_PASS/"         -i custom-config/wrx/workshop-openrc.sh
sed  -e "s/OS_USERNAME=.*/OS_USERNAME=$USR_NAME/"         -i custom-config/wrx/workshop-openrc.sh

# Review
cat    custom-config/wrx/workshop-openrc.sh
# Activate
source custom-config/wrx/workshop-openrc.sh

openstack network list
openstack router  list
openstack server  list
```

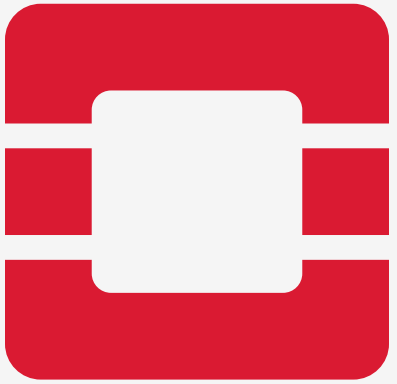
Networking

- Networks:
  - flat
  - vlan
  - vxlan
- Routers
- Floating IPs
- Load Balancers

Networking

External Network

```
# Create external network (VLAN backed)
# as admin
EXT_NET_VLAN=3010
EXT_NET_NAME="shared${EXT_NET_VLAN}"
EXT_NET_CIDR='10.30.10.0/24'
EXT_NET_RANGE='start=10.30.10.192,end=10.30.10.254'
EXT_NET_GATEWAY='10.30.10.1'

openstack network create       ${EXT_NET_NAME}  \
   --provider-physical-network physnet3         \
   --provider-network-type     vlan             \
   --provider-segment          ${EXT_NET_VLAN}  \
   --external

openstack subnet create  ${EXT_NET_NAME}-subnet \
    --allocation-pool    ${EXT_NET_RANGE}       \
    --network            ${EXT_NET_NAME}        \
    --subnet-range       ${EXT_NET_CIDR}        \
    --gateway            ${EXT_NET_GATEWAY}     \
    --no-dhcp
```



Networking

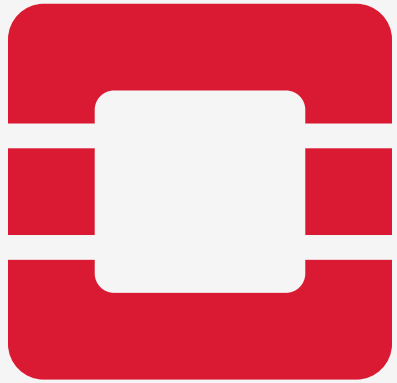
Project Network

```
# Create project network (VXLAN backed)
# as workshop user in workshop project

# Activate
source custom-config/wrx/workshop-openrc.sh

PRJ_NAME=workshop
openstack network create workshop-net

openstack subnet create workshop-subnet \
  --subnet-range 10.10.10.0/24          \
  --network workshop-net                \
  --gateway 10.10.10.1

openstack router create workshop-router
openstack router add subnet workshop-router workshop-subnet
openstack router set --external-gateway shared3010 workshop-router
```

Networking

Virtual Machines

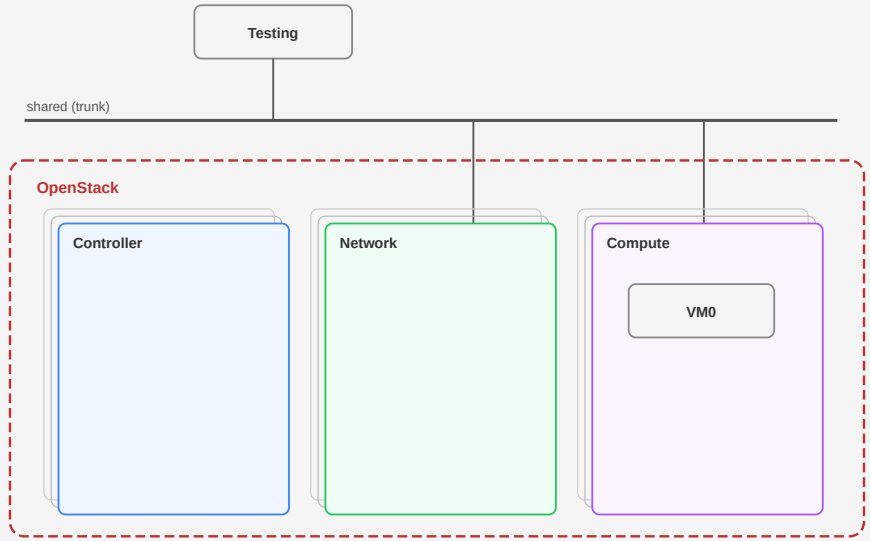
```
# as workshop user in workshop project

IMAGE_NAME=ubuntu-24.04
# IMAGE_NAME=alma-10.1
# IMAGE_NAME=rocky-10.1
# IMAGE_NAME=debian-13

openstack keypair create --public-key ~/.ssh/id_ecdsa.pub mykey

openstack server create          \
  --image    ${IMAGE_NAME}       \
  --flavor   m1.small            \
  --key-name mykey               \
  --nic      net-id=workshop-net \
  vm0

openstack server create                  \
  --image            ${IMAGE_NAME}       \
  --flavor           m1.small            \
  --key-name         mykey               \
  --nic              net-id=workshop-net \
  --boot-from-volume 10                  \
  vm1
```



Networking

How to access?

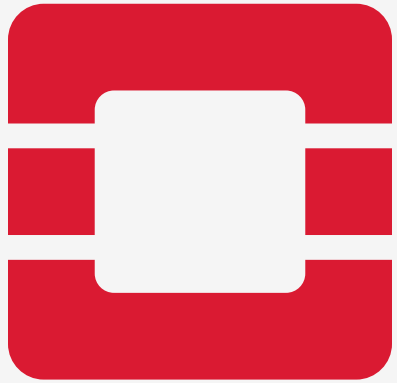
```
# Create a config for a VLAN interface
# on the testing node as root
myid=$(cat /etc/env.json  | jq .id)

cat > /etc/netplan/80-vlan.yaml <<EOF
network:
  version: 2
  vlans:
    xtrn0.3010:
      addresses:
      - "10.30.10.${myid}/24"
      dhcp4: false
      dhcp6: false
      id: 3010
      link: eth3
      link-local: []
EOF

chmod 600 /etc/netplan/80-vlan.yaml

netplan apply
```



Networking

How to access?

```
# Copy the ssh key from deployment to testing node
scp ~/.ssh/id_ecdsa* testing.mgmt.wrx.sckt.net:~/.ssh/

# Assign floating IPs to VMs
openstack floating ip  create shared3010
FIP_VM0=$(
  openstack floating ip  create shared3010 -f value -c floating_ip_address
)
openstack server add floating ip vm0 $FIP_VM0

FIP_VM1=$(
  openstack floating ip  create shared3010 -f value -c floating_ip_address
)
openstack server add floating ip vm1 $FIP_VM1

# Review assigned IPs
openstack server list

# Allow access to vm0
openstack security group create allow-simple-access
openstack security group rule create allow-simple-access \
  --protocol tcp \
  --dst-port 22 \
  --remote-ip 0.0.0.0/0 \
  --ingress
openstack security group rule create allow-simple-access   \
  --protocol icmp \
  --remote-ip 0.0.0.0/0 \
  --ingress
openstack server add security group vm0 allow-simple-access

# You should be able to connect to VMs from the testing node
ping -c 3 <IP>
ssh -i ~/.ssh/id_ecdsa ubuntu@<IP>
```

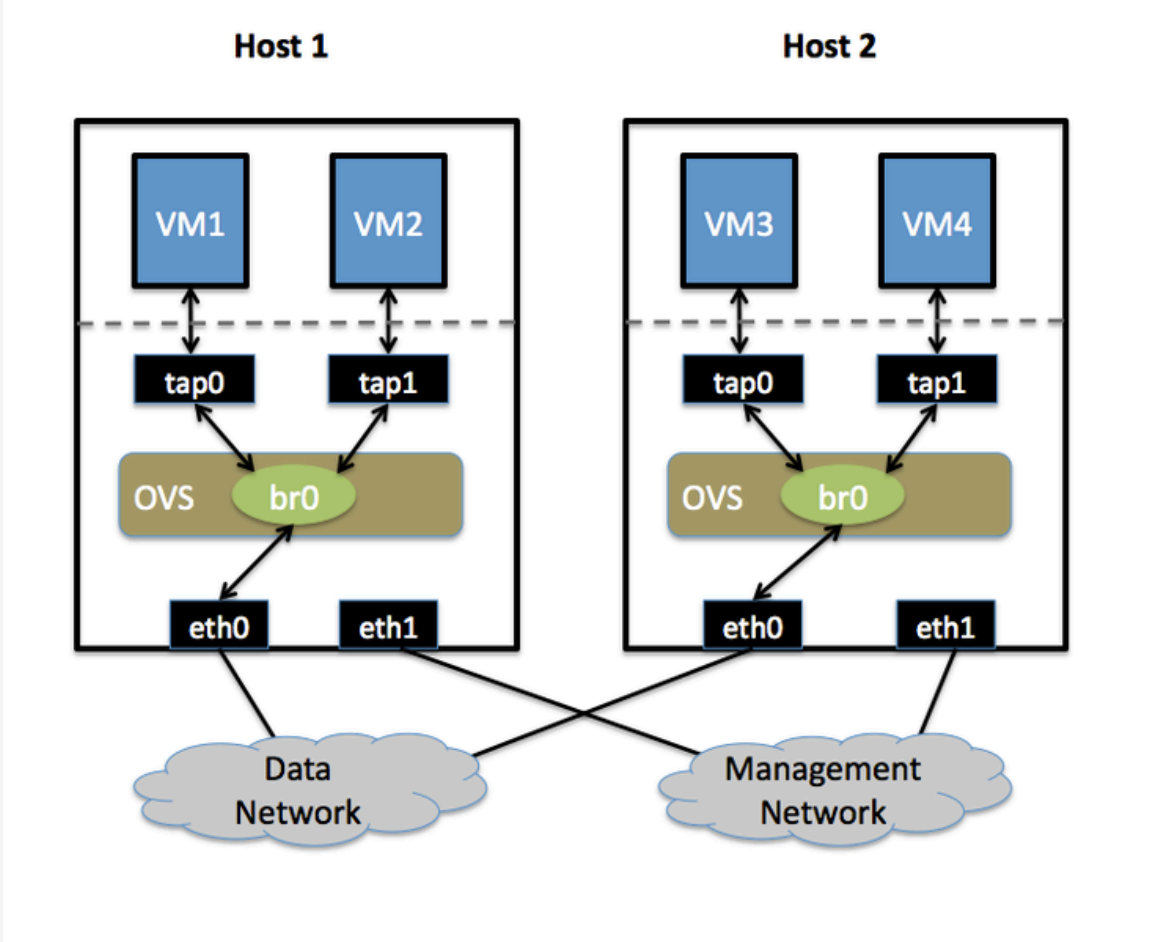


Networking

Debugging

```
# Check VMs and IPs assigned to them
openstack server list --long

# Check the state of network services
openstack network agent list

# Check the router has connectivity to the external network
# Check the router has IPs from the tenant network assigned
openstack router show workshop-router

# On network nodes check if there are network namespaces created for the router
# qrouter-<ROUTER_ID>
# snat-<ROUTER_ID>
# fip-<FLOATING_IP_NETWORK_ID>
ip netns
# Check if IP addresses are assigned
ip netns exec <NETWORK_NAMESPACE_ID> ip a

# On compute nodes check if there are network namespaces created for the router
ip netns
```

Network Namespaces



User Config with Cloud-Init

```
# Create user-data.yaml
cat > user-data.yaml <<EOF
#cloud-config
runcmd:
  - mkdir -p /var/lib/data/test
  - echo "<h1>Hello from write_files</h1>" > /var/lib/data/test/hello.html
fs_setup:
  - device:     /dev/vdb
    filesystem: ext4
    overwrite:  true
mounts:
  - [ /dev/vdb, /var/lib/data, "ext4", "defaults,nofail", "0", "2" ]
write_files:
  - path:        /var/lib/data/index.html
    owner:       root:root
    permissions: '0644'
    content: |
      <h1>Hello from write_files</h1>
EOF
```

User Config with Cloud-Init

```
# Create a new VM with an additional volume
root_uuid=$(
  openstack volume create \
    --image "${IMAGE_NAME}" --size 10 -c Id -f value vm2-root
)
data_uuid=$(openstack volume create --size 10 -c Id -f value vm2-data )

openstack volume list

openstack server create \
  --flavor m1.small \
  --key-name mykey \
  --nic net-id=workshop-net \
  --block-device uuid=$root_uuid,source_type=volume,destination_type=volume,boot_index=0 \
  --block-device uuid=$data_uuid,source_type=volume,destination_type=volume,boot_index=-1 \
  --user-data ./user-data.yaml \
  vm2


# Inspect
openstack server show vm2 -fyaml
openstack volume list

# Create a new floating IP and attach it to the server
vm2_ip=$(openstack floating ip create -f value -c floating_ip_address shared3010)
openstack server add floating ip vm2 $vm2_ip
```

User Config - Inspect the Created VM

```
# Log in from testing node
ssh -i ~/.ssh/id_ecdsa ubuntu@<FLOATING_IP>

cat /etc/fstab
mount | grep vdb

cat /var/lib/data/index.html
cat /var/lib/data/test/hello.html
```

Debugging

```
# Inspect the instance
openstack server show vm2

# Inspect VM with libvirt on the compute host
docker exec -ti nova_libvirt virsh list
docker exec -ti nova_libvirt virsh dumpxml    <INSTANCE_NAME>
docker exec -ti nova_libvirt virsh console    <INSTANCE_NAME> # ctrl+5 to exit
docker exec -ti nova_libvirt virsh domblklist <INSTANCE_NAME>
docker exec -ti nova_libvirt virsh domiflist  <INSTANCE_NAME>

# Inspect volumes backed by Ceph on the compute host
docker exec -ti nova_compute ls /etc/ceph/
docker exec -ti nova_compute ceph -s
docker exec -ti nova_compute ceph --id cinder -s
docker exec -ti nova_compute ceph --id cinder osd pool ls
docker exec -ti nova_compute rbd  --id cinder ls images
docker exec -ti nova_compute rbd  --id cinder ls vms
docker exec -ti nova_compute rbd  --id cinder ls volumes

# Inspect network interface on the compute host
docker exec -ti openvswitch_vswitchd ovs-vsctl show

# Within the instance
cloud-init status
cloud-init status --long
```

Block Devices

```
# How many VMs are running? How many volumes should we see?
openstack volume list

# Create a new volume and attach it
openstack volume create  --size 1 vm1-data
openstack server add volume   vm1 vm1-data
openstack volume attachment list

# Inspect it within the VM from the testing node
lsblk

# Create some data on it
mkfs.ext4 /dev/vdb
echo "$(date) Hello from vm1"

# Take a snapshot on the deployment node
openstack volume snapshot create --volume vm1-data vm1-data-snap1 [--force]
openstack volume snapshot list

# Change the data
echo "Let us add a new line into the file after the snapshot was created"  >> /mnt/test
cat  /mnt/test
```

Block Devices

Revert a Volume

```
# Restore from the snapshot
openstack server stop   vm1
openstack server list;  openstack volume list;
openstack volume snapshot list --volume vm1-data

openstack server remove volume vm1 vm1-data
openstack volume revert            vm1-data-snap1
openstack server add    volume vm1 vm1-data
openstack server start         vm1
```

Block Devices

More Operations

```
# Create a volume from an image
openstack volume create --image cirros --size 1 vm1-cirros

# Create a volume from a snapshot
openstack volume snapshot create --volume vm2-root vm2-root-snap1 [--force]
openstack volume create --snapshot vm2-root-snap1 ubuntu-24.04-from-vm2

# Create an image from a volume
openstack image create  --volume ubuntu-24.04-from-vm2 ubuntu-24.04-custom
openstack image list
```

Resource Segregation

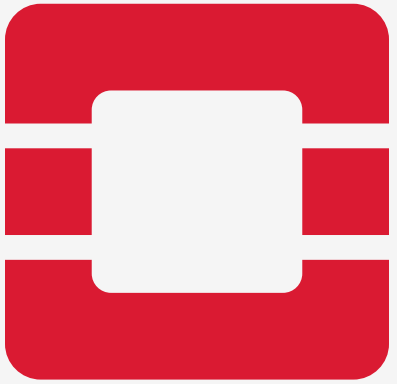
- Projects
- Host Aggregates
- Volume Backends
- Network Agents
- Availability Zones

Availability Zones

is a human-readable label that represents a **physically isolated slice of data center infrastructure**. It is the primary tool used to group hardware into separate **fault domains** so that a single physical failure does not take down an entire cloud application.

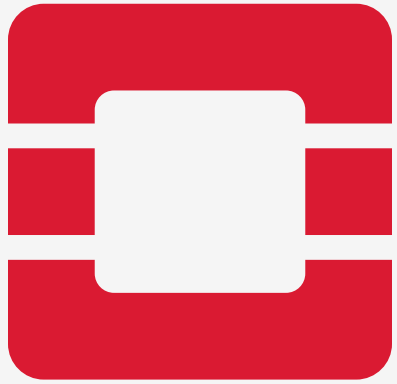
Availability Zones

- Power Distribution Unit
- Data Center (Room)
- Cooling Domain
- Top-of-Rack (ToR) Switch Domain
- Storage Backend



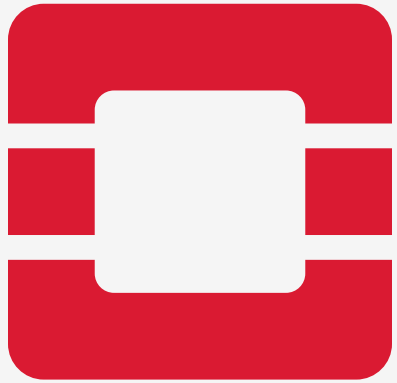
Availability Zones

```
openstack aggregate create --zone az1 prd-az1
openstack aggregate create --zone az2 prd-az2

openstack aggregate add host prd-az1 compute01
openstack aggregate add host prd-az1 compute02
openstack aggregate add host prd-az1 compute03

openstack aggregate show prd-az1
openstack aggregate show prd-az2

openstack availability zone list --compute
openstack availability zone list --network
openstack availability zone list --volume
```

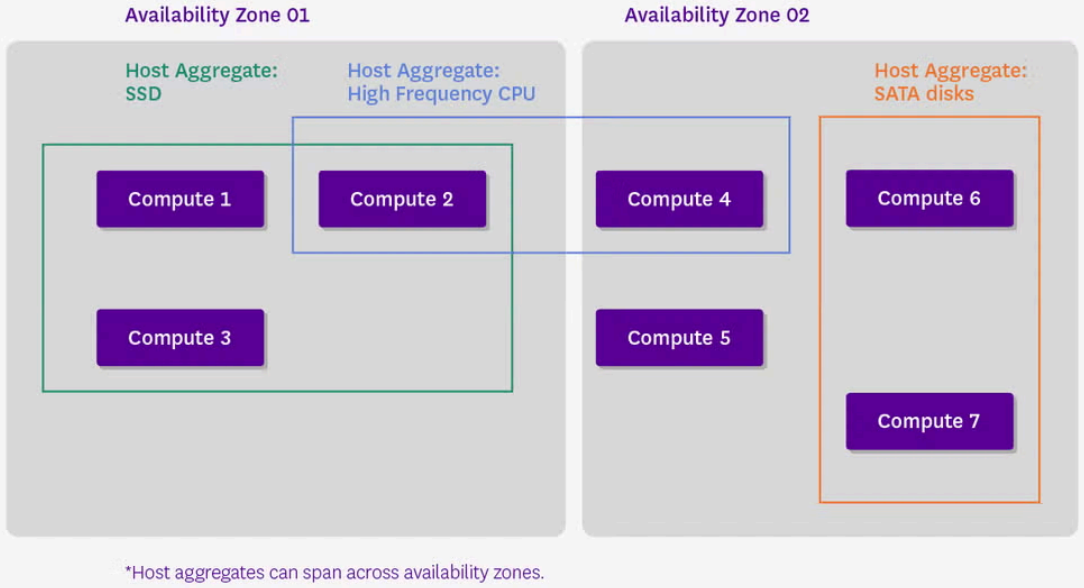


Host Aggregates

```
openstack aggregate create hw-type-A
openstack aggregate create hw-type-B
openstack aggregate create windows
openstack aggregate create gpu

openstack aggregate add host hw-type-A compute01
openstack aggregate add host hw-type-A compute02
openstack aggregate add host hw-type-A compute03
openstack aggregate add host hw-type-B compute11

openstack aggregate add host windows compute01
openstack aggregate add host windows compute11

openstack aggregate add host gpu compute11

openstack aggregate show windows
```



Host Aggregates vs. Availability Zones



Links

- OpenStack Docs
- OpenStack Configuration Guides
- OpenStack Releases
- Cloud-Init Docs